

2.2 A Quad-Chiplet AI SoC with Full-Chip Scalable Mesh Over 16Gb/s UCle-Advanced Die-to-Die Interface for Large-Scale AI Inferencing

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Abstract

A 4nm-based quad-chiplet with an advanced packaged LLM accelerator achieving 56.8TPS on LLaMa v3.3 70B with single-batch 2k/2k input/output sequences. The architecture combines chiplet-based design, low-latency die-to-die interfaces, unified mixed-precision

compute, holistic synchronization, and HBM3E with advanced power schemes to sustain bandwidth, capacity and thermal stability. The SoC integrates four NPU chiplets, four HBM3E modules, and four ISCs for stable power delivery.

This paper presents an NPU architecture optimized for rapidly evolving, large-scale LLMs such as GPT-4.5, Claude 4, Gemini 2.5, and Llama 4, addressing two primary computational challenges. The compute-bound prefill phase requires a massive amount of floating-point operations, exacerbated by increasing model complexity and prompt engineering, while the memory-bound decoding phase is limited by KV cache bandwidth, becoming more severe with context windows extending to millions of tokens. To overcome these challenges, we propose a scalable chiplet-based architecture that defines the minimum system granularity, enabling the construction of a large, virtually monolithic system. Modular chiplets support both scale-up and scale-out expansion, allowing growth from single-die setups to multi-chiplet clusters. The design also accommodates I/O chiplets for seamless integration of additional compute or memory resources, mitigating compute and memory bottlenecks while establishing a robust foundation for next-generation LLMs.

We present a quad-chiplet configuration as the foundational building block, delivering 56.8TPS on Llama 3.3 70B [1] with a single-batch 2k/2k input/output sequence in a single-chip setup. The architecture combines chiplet-based design, a low-latency die-to-die interface with memory load-store semantics, and a unified compute and interconnect fabric that scales efficiently across chiplet and chip boundaries. Holistic synchronization ensures high utilization across model phases, while HBM3E integration with advanced power schemes sustains bandwidth, capacity and thermal stability. Together, these innovations establish a robust platform that not only addresses the continuous rise in compute density and design scale but also provides the scalability required for future system expansion.

The AI SoC integrates four identical NPU chiplets, four HBM3E modules, and four integrated silicon capacitors (ISCs), as illustrated in Figs. 2.2.1 and 2.2.7. Each NPU chiplet employs three UCle channels, with topology-aware die sliding and rotation applied to ensure seamless vertical and horizontal mesh continuity. The ISCs feature multiple power domains, actively supplying sufficient capacitance to both the NPU and HBM3E.

The SoC interfaces with the host system via two PCIe Gen5 $\times 16$ links, supporting SRIOV and peer-to-peer connectivity. A command processor compiles descriptors and manages task scheduling both cooperatively and independently across all NPU chiplets, in conjunction with a full-chip synchronization manager and DMA engines. The SoC delivers 2PFLOPS (FP8) of compute capability through a unified multiple- and mixed-precision arithmetic unit [2]. By eliminating the need for separate arithmetic units across different precision levels, the design improves compute density while supporting per-operand configurability for diverse precisions, enabling resource-efficient mixed-precision execution. Complementing this, the SoC integrates 512MB of on-chip SRAM organized in a multi-cycle, multi-bank architecture: 256MB high-density scratchpad memory providing 128TB/s aggregated bandwidth, and 256MB of shared memory delivering 64TB/s aggregated bandwidth. This balanced memory hierarchy supports both low-latency buffering and high-bandwidth shared access, sustaining throughput under demanding workloads. The combined compute and memory resources are defined as a Neural Core, grouped into Neural Core Clusters. These clusters communicate across the full-chip interconnect, which extends seamlessly over chiplet boundaries via UCle-based links. Moreover, the architecture is explicitly designed to enable scalability beyond package boundaries, providing a pathway to future system expansion without sacrificing performance efficiency.

The data transfer across all chiplets is orchestrated through four key architectural components shown in Figs. 2.2.1 and 2.2.2: a full-chip spanning mesh network, a configurable multi-engine DMA, a synchronization mechanism, and a proprietary die-to-die interface. Each chiplet integrates 16 Neural Cores and 64MB of shared memory, organized into an 8×4 granular mesh interconnected via 64 routers and supported by a DMA subsystem with 8 execution engines. Leveraging the die-to-die interface, the system seamlessly extends into a unified virtual mesh topology, scaling to 256 routers across the entire multi-chiplet configuration.

To dynamically adapt to diverse AI workloads, traffic-aware prioritization and QoS controls govern real-time resource allocation. Fully interleaved shared-memory access paths and multi-path routing deliver up to $4 \times$ peak per-node bandwidth, while a virtual synchronization network embedded in the mesh minimizes dependency resolution latency, as illustrated in

the green path of Fig. 2.2.2. In clustered configurations, groups of eight Neural Cores and 32MB of shared memory are connected through a customized mesh NoC comprising three logically independent channels (D, R, and C). The D-channel handles bulk data transfers, while the R- and C-channels serve requests, responses, and control messages. Routers employ XY-routing with turn-model constraints and weighted round-robin arbitration, where adjustable weights enable fine-grained QoS tuning from a system-wide perspective. Distributed shared memory is partitioned into 16 slices across NoC nodes, with logical mapping managed through address-decode tables (ADT) in the network interface unit (NIU) for efficient interleaving. To improve yield and manage design complexity associated with increasing die and chip size, a harvesting scheme is supported at both the cores and shared memory levels, with the mesh isolating defective nodes from the system, as illustrated in the red-dashed region of Fig. 2.2.2.

The multi-engine DMA exploits both local/remote HBM3E and shared memory, sustaining up to 2.6TB/s per-DMA bandwidth. Per-task QoS features further reduce long-tail latency and mitigate dependency bottlenecks, streamlining execution across heterogeneous workloads. Meanwhile, the synchronization mechanism operates through a dedicated virtual channel over the mesh, eliminating direct peer-to-peer dependencies between Neural Cores, DMAs and synchronization managers. Each chiplet incorporates a synchronization manager with hardwired sync control for scalability over the previous scheme [3], coordinating cross-chiplet operations either centrally or autonomously to minimize inter-entity interactions and enhance efficiency across chiplet boundaries.

The chiplet interface employs a UCle-based latency-optimized protocol, achieving 11ns FDI-to-FDI latency at 16Gbps, while fully utilizing the available port bandwidth. By extending load-store memory semantics across dies, the interconnect enables the multi-chip system to operate as a virtually monolithic unit with inherent support for both scale-up and scale-out expansion. To ensure reliability in data center environments, the architecture incorporates a configurable switching that trades marginal performance for significant improvements in MTBF and MTTF, thereby reducing downtime and lowering MTTR to enhance overall system availability. Testability, often constrained in multi-die systems, is improved through multi-layer loopback paths that extend beyond the UCle specification. These include transaction-level and channel-level modes, as well as inbound-to-outbound loopback with optional address remapping, offering comprehensive diagnostic coverage. As shown in Fig. 2.2.5, bathtub and eye-diagram measurements confirm high signal fidelity and wide margins, with error-free operation verified at 16Gbps during the bring-up phase. Remarkably, these results were achieved within two months of silicon arrival, demonstrating the robustness and maturity of the chiplet interface design.

With a thermal design power (TDP) of 600W, the system encounters transient surges exceeding twice the nominal level, posing significant challenges for power integrity (PI). The lock-step activation and deactivation of multiple Neural Cores, while essential for functional correctness, induces large transient currents (high di/dt) that trigger voltage droops and threaten stability. To address this, we employ a hardware-based staggering technique, skewing the start-up sequences of individual Neural Cores. As shown in Fig. 2.2.4, this approach markedly reduces peak transient current: the synchronous method (top-left) exhibits steep current slopes and severe voltage noise, whereas the staggered method (top-middle) achieves gentler slopes and improved supply stability. In addition, power throttling and smoothing mechanisms, applied at both the Neural Core and inter-chiplet granularity, dynamically regulate instruction execution rate and issue count over a moving time window to further suppress transient fluctuations. In parallel, the high-transient HBM3E traffic, comparable in intensity to NPU compute bursts, places additional stress on the power delivery network. To address this, we integrate a ISC die, embedding distributed on-chip capacitance across the V_{DD} rails of both the HBM3E and PHY. This reinforcement effectively suppresses traffic-induced voltage fluctuations, as demonstrated in Fig. 2.2.5, where impedance profiles with ISC integration show a clear reduction in resonance peaks versus the baseline. Through these combined architectural and physical design measures, we achieved stable HBM3E operation, validating both EOM (UI) and performance metrics within just three weeks of silicon bring-up. Collectively, these innovations establish a robust PI solution, enabling reliable operation under extreme transient workloads.

Our architectural innovations establish a petaflop-class LLM solution that not only matches but surpasses the capabilities of leading commercial accelerators. Figure 2.2.6 presents the measurements during bring-up up to 1.7 $\times$ higher power efficiency compared to an H200 [5], and emulator-correlated performance estimations at the design-target configuration indicate highly promising efficiency gains, further underscoring the competitiveness of our approach. These results demonstrate the practicality of our approach in delivering scalable, high-efficiency AI acceleration that can serve as a compelling alternative to existing GPU-based solutions. This design serves as a foundational building block for larger AI systems, supporting both scale-up and scale-out through chiplet expansion. To further enhance scalability, we are developing I/O and memory expander chiplets as shown in Fig. 2.2.7 connected through the scalable mesh over the die-to-die interfaces. This scalable architecture supports cross-node/rack LLM systems for trillion-parameter models with million-token contexts, reshaping large-scale AI economics.

Acknowledgement:

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References:

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Figure 2.2.1: Full-chip block diagram with four NPU chiplets, four HBM3E modules and four ISCs.

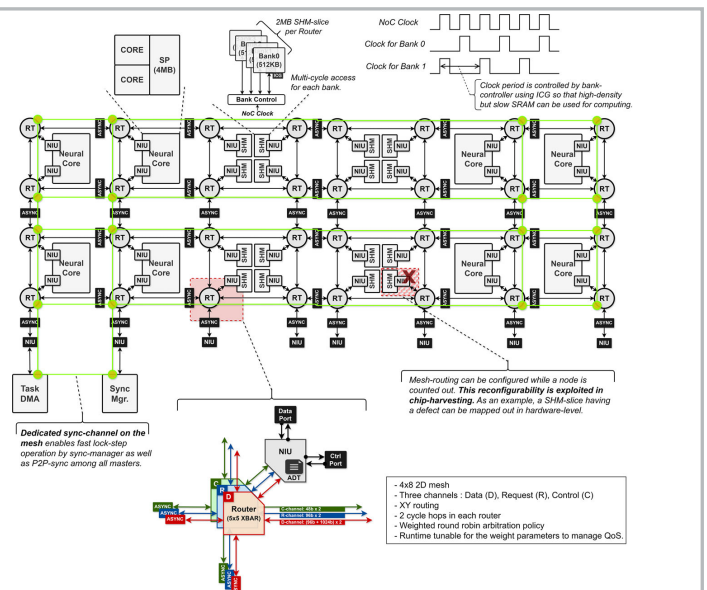


Figure 2.2.2: Full-chip scale mesh with neural cores, DMA, synchronization manager connected with three logically independent channels.

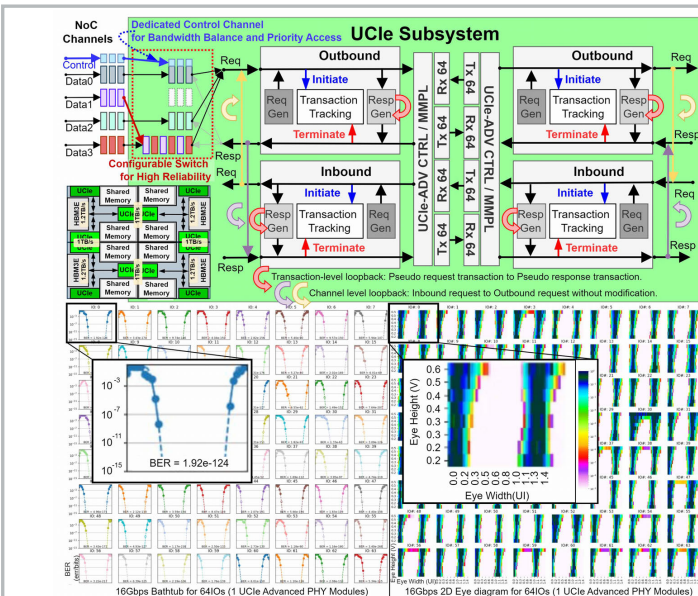


Figure 2.2.3: Chiplet interface subsystem with advanced UCle die-to-die links. Bathtub and eye diagram snapshots at target speed during silicon bring-up.

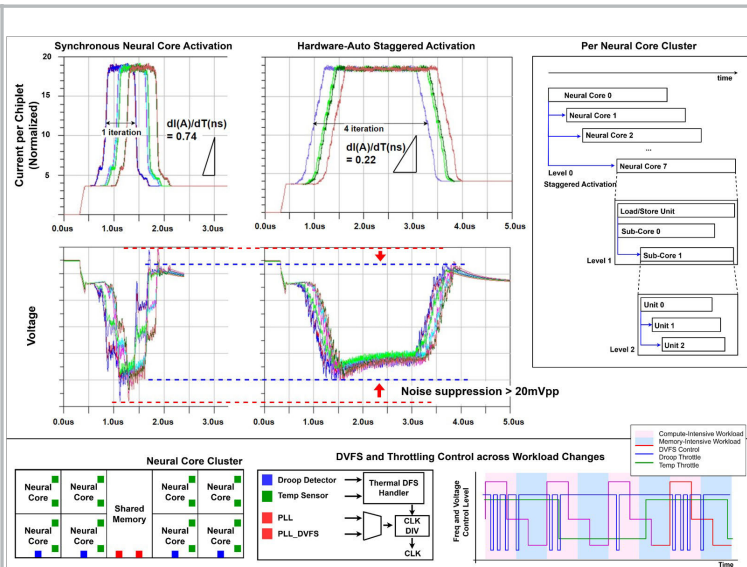


Figure 2.2.4: Hardware-based staggering technique and core-group closed-loop control for voltage droop and temperature, with per-cluster DVFS control loop.

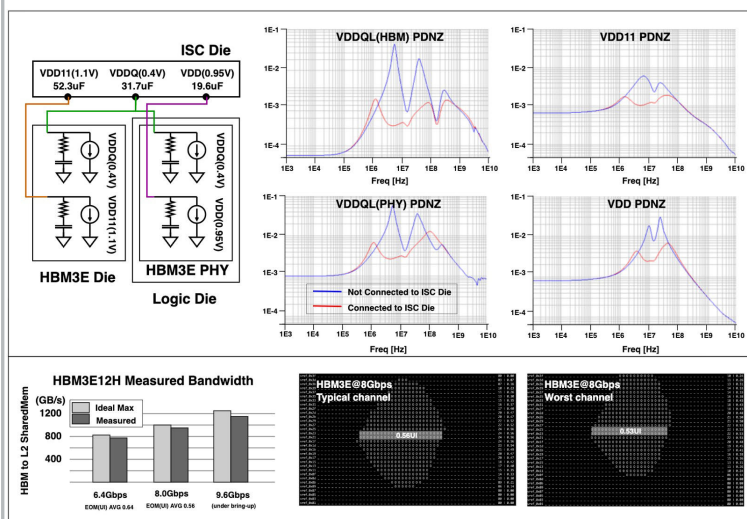


Figure 2.2.5: ISC for power integrity enhancement of HBM3E operation. Measured bandwidth performance and eye diagram; the eye diagram at the target speed (9.6Gbps) will be updated after silicon bring-up.

	H100 SXM [4]	H200 SXM [5]	This work
Process node	5nm	5nm	4nm
Native compute capabilities (TFLOPs)	fp16: 989.5 fp8: 1979	989.5 1979	1024 2048
On-chip SRAM (MB)	83 (L1+L2)	83 (L1+L2)	512
Chip (Package) TDP (W)	700	700	600
DRAM capacity (GB)	80	141	144
DRAM BW (TB/s)	3.35	4.8	4.8

	Input Length	Output Length	H100 SXM [4]	H200 SXM [5]	This work Silicon measured (bring-up config)
Performance (TPS)	512	512	33.6	42.1	38.9
	512	2048	32.7	42.3	39.4
	2048	512	32.5	40.5	36.9
	2048	2048	33.2	41.6	38.4
Power (W)	512	512	338.2	505.1	276.4
	2048	512	337.9	504.3	274.2
	2048	512	337.4	505.3	276.7
	2048	2048	337.7	505.8	275.9
TPS/W	512	512	0.099	0.083	0.141
	512	2048	0.100	0.084	0.144
	2048	512	0.096	0.090	0.133
	2048	2048	0.098	0.082	0.139

Silicon measurement results @ Evaluation board (EVB)
(For H100/H200, taking 85% of the measured power, chip-only power out of board)
Please note this refers to actual application running power, not TDP.

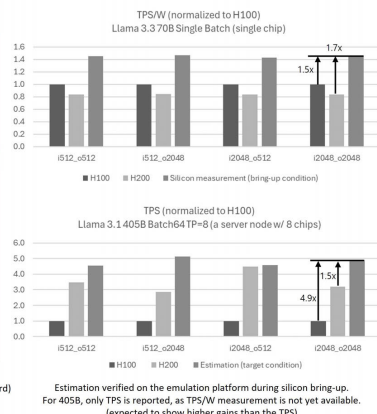


Figure 2.2.6: Performance (TPS) and power efficiency (TPS/W) comparison on silicon vs. flagship AI processors. Weight-only quantization applied due to DRAM limits on the 70B model.

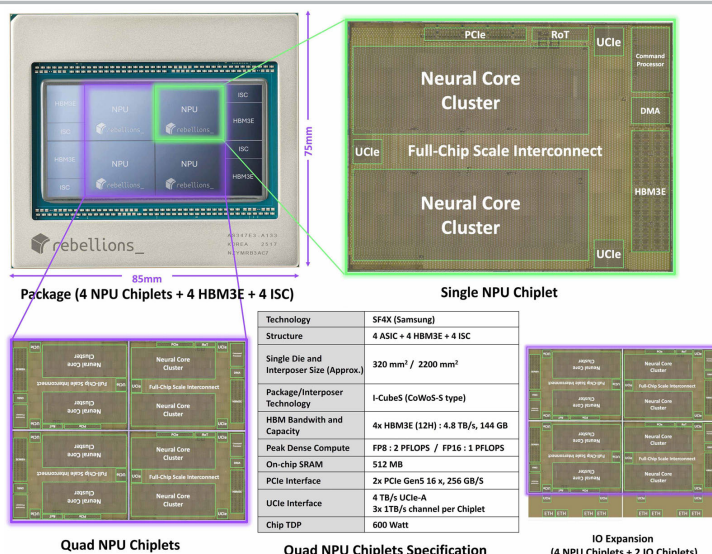


Figure 2.2.7: Package photo with four NPU chiplets, four HBM3E modules and four ISC dies. Various package configurations are being developed in parallel with the NPU chiplet.